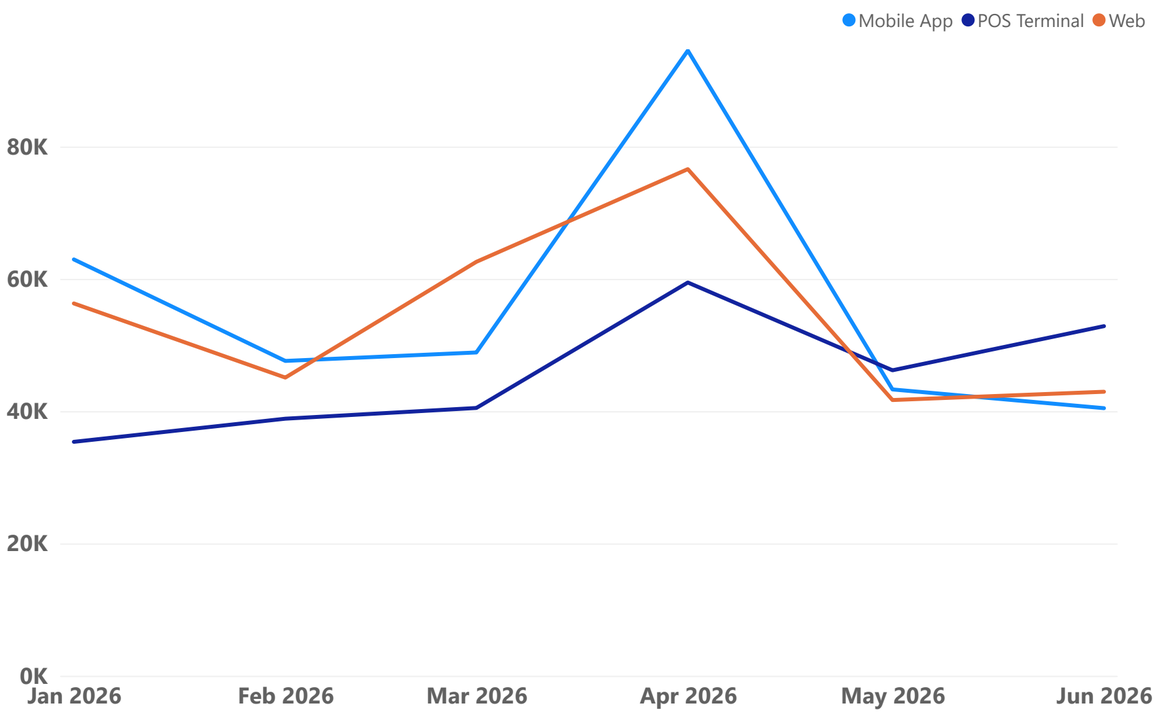
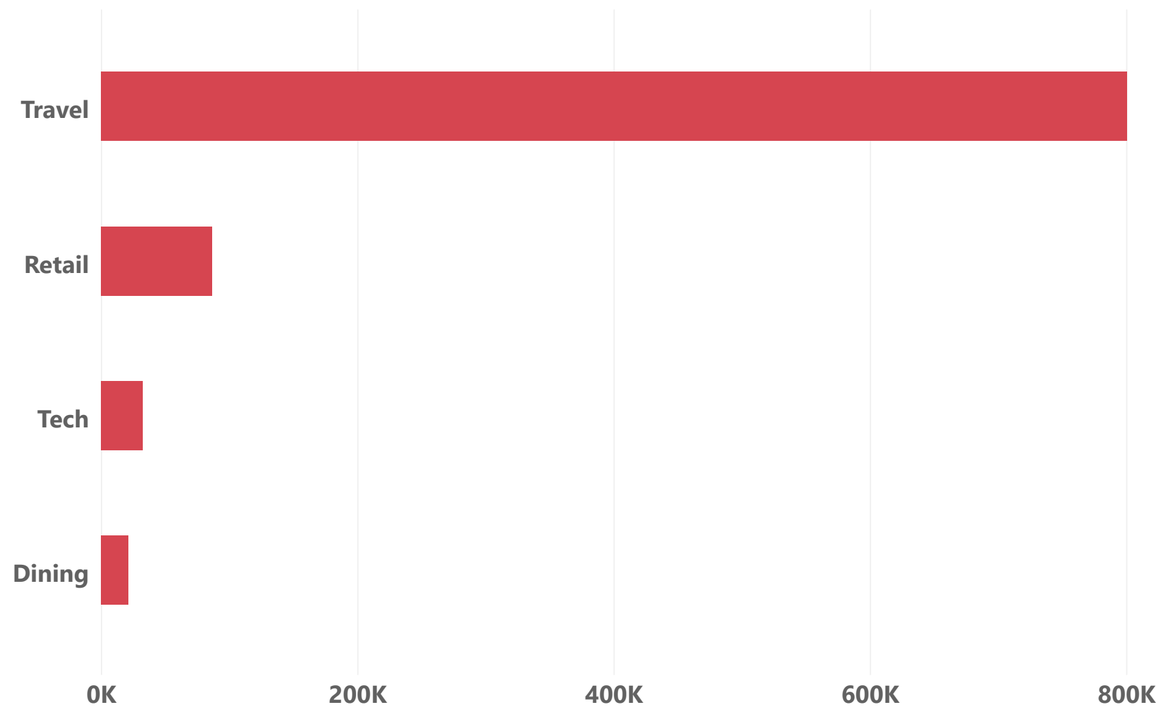
Executive Fraud Overview

Total Transactions	Fraud Rate	Fraud Amount	Confirmed Fraud Investigations	Total Investigations
50,000	1.00%	945,302	450	2,925

Fraud Amount by Year, Month & Channel



Fraud Amount by Industry (mapped merchants only)



April Mobile app peak is driven by one 43,005 transaction

Key Findings

- Travel transactions are about 8% fraud while others have fraud rate of below 1%.
- Fraud rate for amount 1000+ is above 16%, against about 0.5% for other transaction amounts.
- The rule "Night-Time-Activity" raised 2,870 alerts and only 53 (1.8%) were real fraud.
- The model catches 90% of fraud, but only 15.38% of its flags are real fraud, so analysts review about 6.5 transactions for every fraud they find.

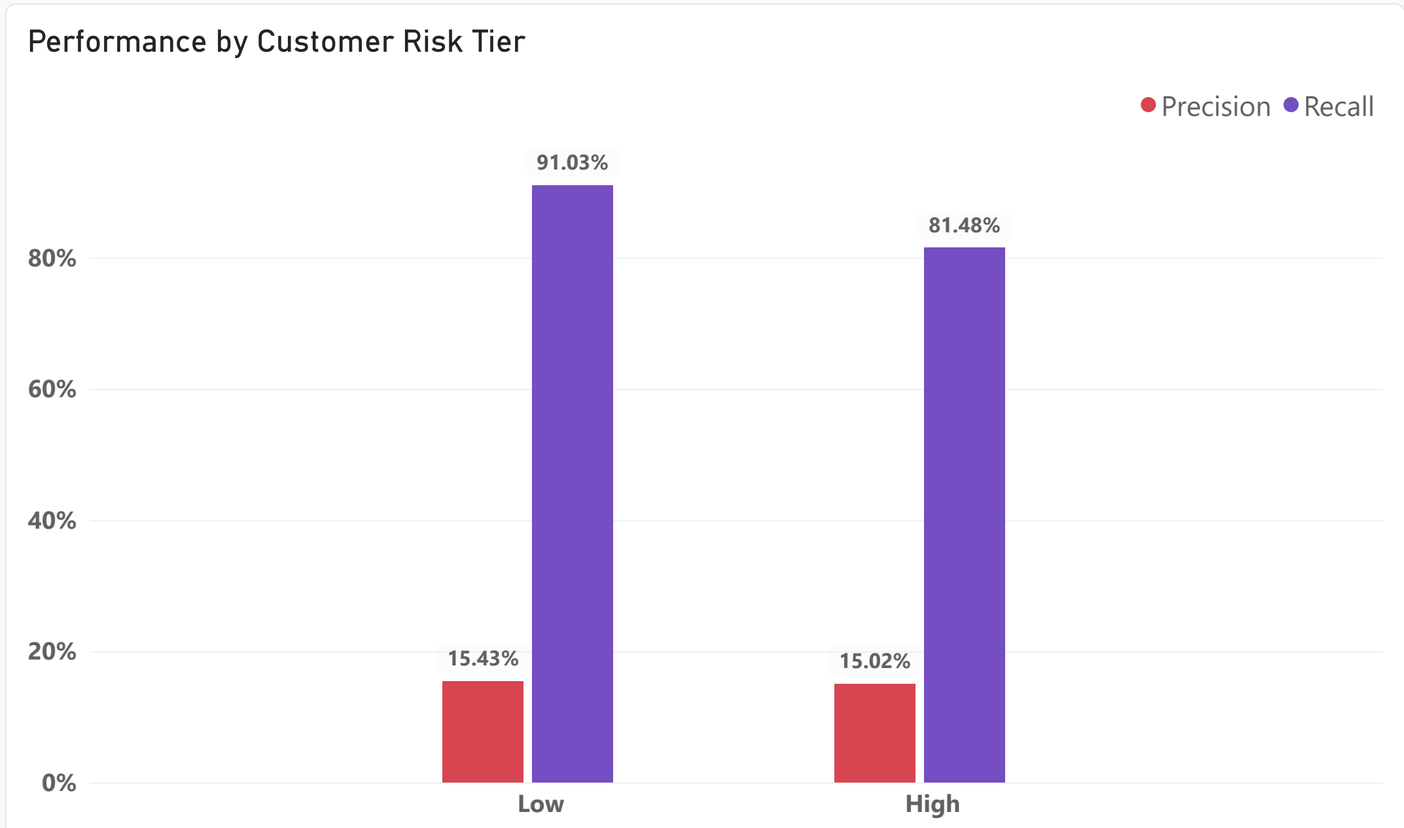
Detection Performance

Precision	False Positive Rate	Recall
15.38%	5.00%	90.00%

Confusion Matrix

Actual	High_Risk - ML Flag	Nominal
False	2,475	47,025
True	450	50
Total	**2,925**	**47,075**

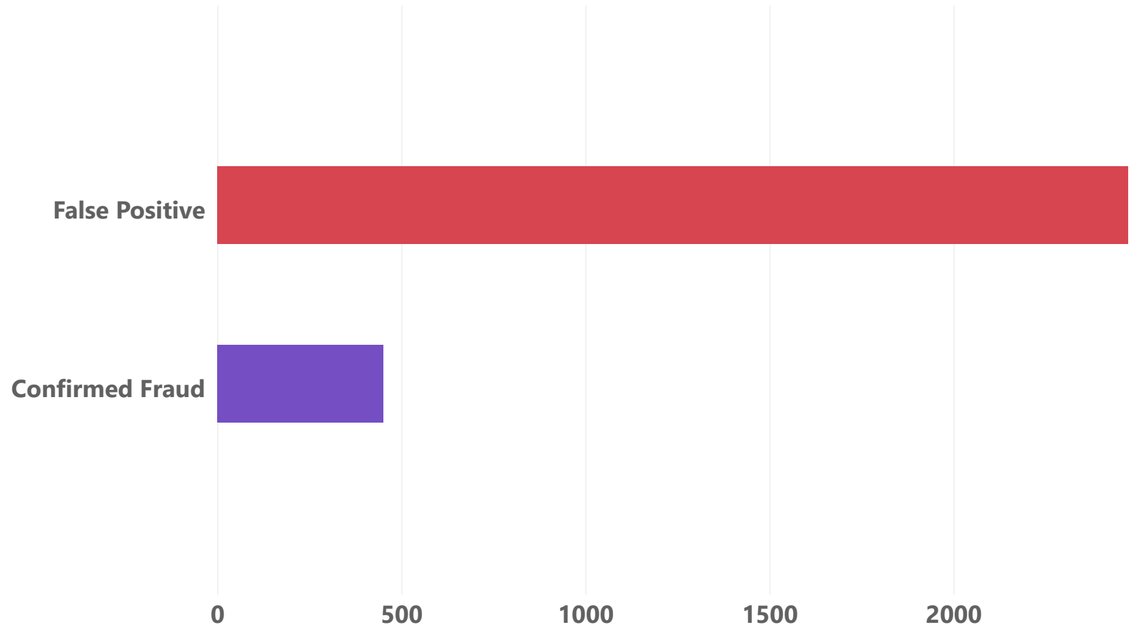
Recall looks lower for High-risk customers, based on 54 fraud cases



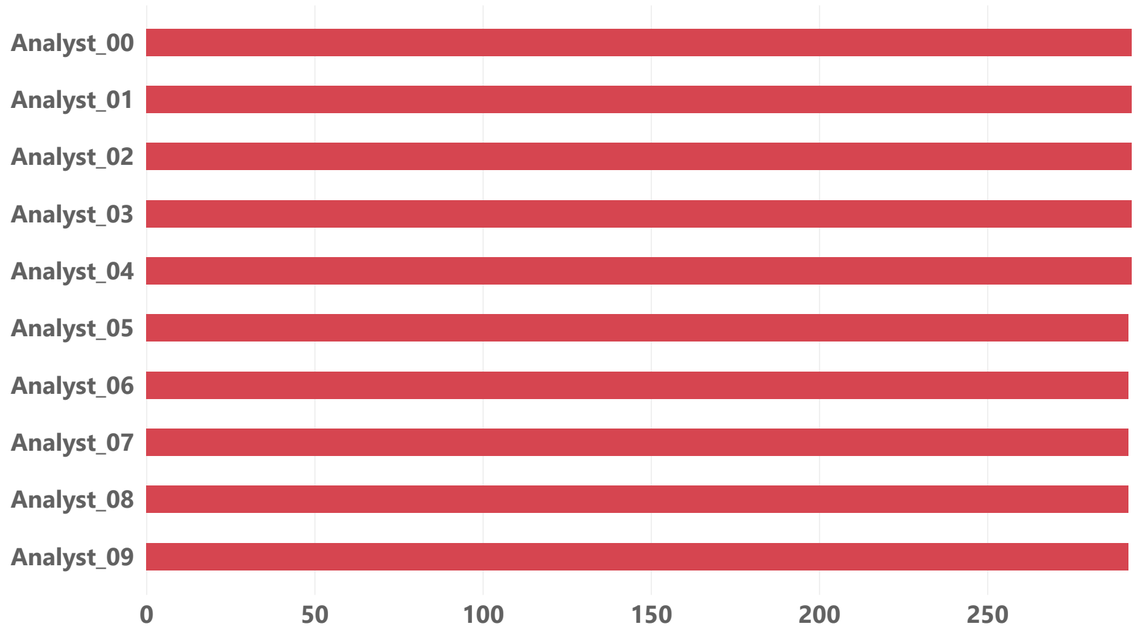
Investigation Operations

False Positive Investigations	Average Case Turnaround Hours	Confirmed Fraud Investigations	Total Investigations
2,475	47.97	450	2,925

Investigations by Outcome



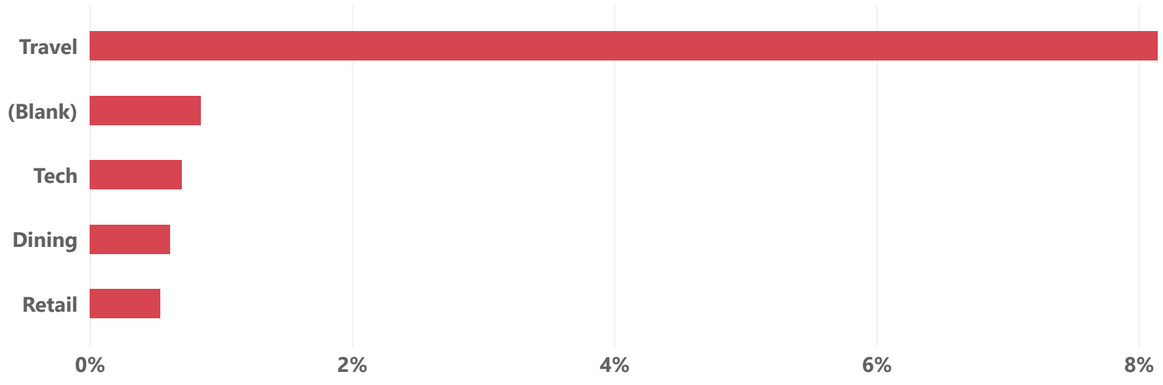
Cases per Analyst



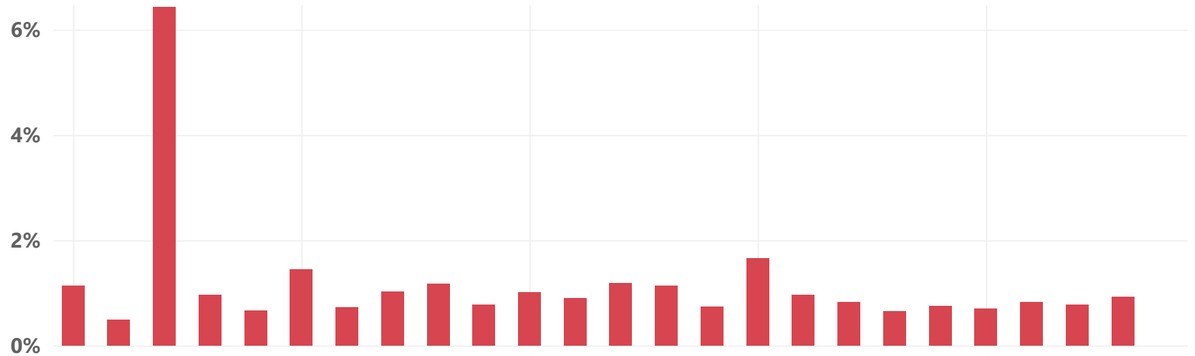
Data note: all 2,925 cases are closed, and cases are spread evenly across analysts. This is synthetic data, so no analyst performance ranking is drawn.

Fraud Pattern Analysis: where fraud concentrates

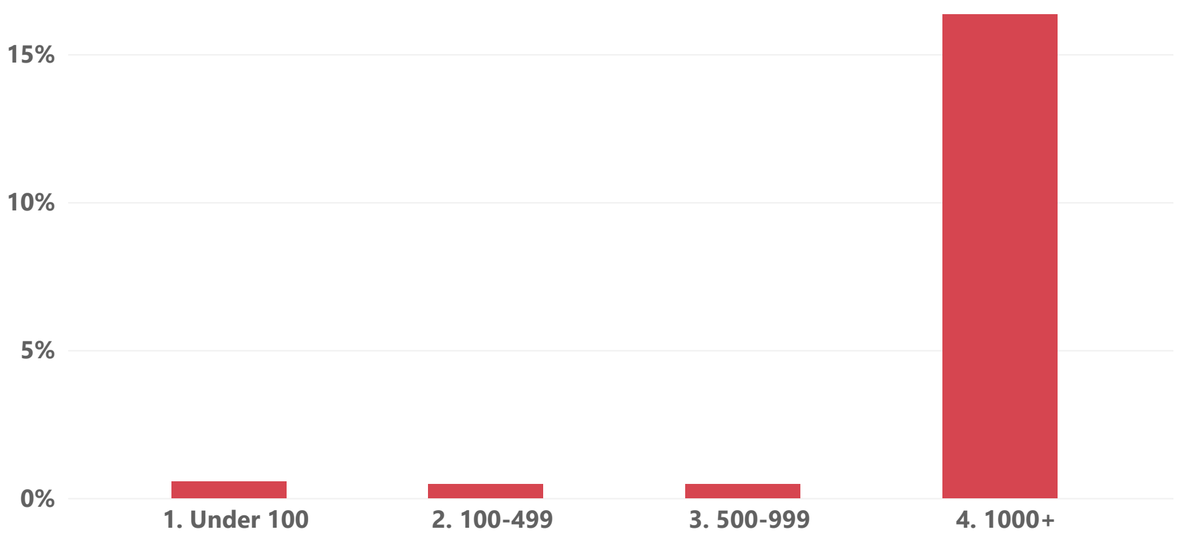
Fraud Rate by Merchant Industry



Fraud Rate by Hour of Day



Fraud Rate by Transaction Size



Rule Effectiveness Table

Rule Name	Alert Volume	Fraud Alerts	Alert Hit Rate
Model-Score-Threshold	2925	450	15.4%
Night-Time-Activity	2870	53	1.8%
Multiple-Risk-Signals	661	426	64.4%
Large-Tx	610	208	34.1%
High-Risk-Merchant	223	199	89.2%
Geo-Mismatch	200	200	100.0%
New-Device	200	200	100.0%
Cross-Region-Activity	81	39	48.1%
Rapid-Amount-Escalation	30	30	100.0%
Velocity-High	20	20	100.0%
Total	**7820**	**1825**	**23.3%**

Limitations and Data Notes

- The data is synthetic, so findings don't describe real banks.
- Only about 1% of transactions are fraud, so precision and recall depend heavily on a small group.
- Investigation outcomes match the fraud labels exactly, so the confirmation rate is not independent proof of model quality.
- All 2,925 investigations are closed, so queue size over time can't be measured.
- No currency, recovery, geography, or churn data exists, so loss recovery and regional analysis weren't attempted.
- Some patterns, such as the 2am spike and the High-risk tier results, rest on small groups of fraud cases.
- e 2 fraud transactions (4,591.65) have no merchant.